

Internship Report

On

State Data Centre Policy Index (SDPCI): A 33-Incentive Comparative Assessment of Data Centre Policies Across 17 Indian States

By

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Preface

This report presents a comparative assessment of state-level Data Centre policies in India, developed during an internship with the Research & Networking Vertical at NITI Aayog. Data centres have become a focal point of state industrial policy in the last five years, and states have responded with a wide range of fiscal, regulatory and infrastructural incentives aimed at attracting hyperscale investment. This report builds a common, points-based scoring instrument, the State Data Centre Policy Index (SDCPI), to compare these policies on a single measurable scale, and tests how far the resulting ranking depends on the relative importance given to different categories of incentive. It is intended to support the Vertical's engagement with states on data centre investment facilitation.

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1. Executive Summary

1.1 Brief Overview of the Assignment

This report was prepared during an internship with the Research & Networking Vertical, NITI Aayog. The assignment aims to build a common, points-based framework to compare the Data Centre policies notified by Indian states, and to test how robust the resulting comparison is. The study covers 17 states with a publicly available Data Centre policy, or with explicit Data Centre provisions inside a broader industrial or IT policy.

Each state was scored against a 33-incentive Scorecard, with a maximum of 132 points, grouped into five policy dimensions: Fiscal / Financial Incentives, Regulatory / Procedural Incentives, Infrastructure Assurance, Sustainability and Green Incentives, and Emerging / Other Incentives. Scoring follows a strict Data-Centre-applicability rule, under which an incentive is credited only where the published policy text explicitly extends it to Data Centres. A 20,000-trial Monte Carlo simulation was then used to test how sensitive the resulting state ranking is to the weight given to each of the five dimensions.

1.2 Key Findings

The Monte Carlo results show that states competing mainly on the size of their fiscal package face the greatest rank instability. Chhattisgarh falls six ranks, from sixth on raw score to twelfth on the Composite Policy Index, because its Fiscal / Financial strength (35/60) is not matched by Infrastructure Assurance (0/12) or Sustainability (2/24). Rajasthan's balanced profile across all five dimensions is what makes it the only state holding a Top 4 position in all 20,000 trials. States should treat dimensional balance, not incentive size alone, as the design objective.

Nine of the seventeen states studied are penalised most heavily by weak or absent commitments on assured power, dual-grid connectivity and assured water supply. This dimension also shows the widest correlation range of any in the panel. Explicit, quantified commitments in this area, rather than general infrastructure language, would address the most common weakness identified in this study.

Infrastructure Assurance, covering assured power, dual-grid connectivity and assured water supply, is the most common weak point nationally. It is the dimension most strongly associated with a worse rank for nine of the seventeen states in the panel.

The wide variation in how comprehensively states address each policy dimension points to the absence of a shared baseline. A model framework that requires states to address fiscal incentives, regulatory procedures, infrastructure, sustainability and emerging technology, while leaving the scale of each incentive to individual states, would raise the overall standard without limiting competitive differentiation. This is a reasonable next step suggested by the findings, though the report does not directly model or test its effect.

The strict Data-Centre-applicability rule materially lowers the score of states whose general IT/ITeS incentives are generous but not explicitly extended to Data Centres, such as Punjab and Jharkhand, on several individual incentives.

The strict applicability rule demonstrates that generous incentives in broader IT or industrial policies cannot automatically be assumed to apply to data centres. States should therefore explicitly specify which incentives are available to data centre projects, reducing ambiguity and ensuring that investors can clearly identify the benefits they are eligible to claim.

Some of the lowest scoring states, Kerala in particular, describe an intention to support Data Centre investment without specifying subsidy amounts, power tariffs, land terms or approval timelines. A policy that states an amount, a duration and a clear eligibility condition is more useful to an investor and more comparable across states than one that states only an intention.

Sustainability is comparatively underdeveloped across most of the states studied. Given the scale of power demand associated with Data Centre growth, treating renewable energy access, green building standards and water efficient cooling as central provisions, rather than optional additions, would align policy design with the sector's own resource needs.

2. Introduction

2.1 Background and Context

Data Centres have been a highly dynamic topic in India's policy space in the recent past. Eleven out of twelve states that have a publicly available policy on Data Centres has either been made or extensively amended after 2020. This surge has been fuelled by the rising demand of hyperscale data centres against the backdrop of the global AI surge. The country's total data centre capacity has increased from about 375 MW in 2020 to around 1,500 MW by 2025, a growth of roughly 300 percent (PIB, 2026).

This growth is primarily influenced by state-level Data Centre policies. As the attention of data centre builders shifts towards India, states have been competing to attract such companies to their geographical territory through policies that provide the maximum possible incentives and subsidies.

Over the past decade, several states have moved from treating data centres as a subset of general IT/ITeS policy toward publishing dedicated data centre policies with attractive fiscal and infrastructural incentives. Telangana's 2016 policy was among the earliest dedicated frameworks, followed by Tamil Nadu, West Bengal and Uttar Pradesh in 2021, Karnataka, Haryana and Odisha in 2022, Rajasthan in 2025, and most recently Gujarat in 2026. This pattern of states independently designing overlapping incentive packages to attract the same pool of investment is a textbook example of competitive federalism in India. The recent disagreement between ministers from Karnataka and Andhra Pradesh over a large technology company's decision to locate in Andhra Pradesh rather than Karnataka illustrates the intensity of this competition. It raises a question central to fiscal federalism theory: does inter-state competition for data centre investment improve overall policy quality, or does it risk a race to the bottom in which states compromise their own fiscal base to match rivals' concessions?

Despite this growth in Data Centre policies, no studies were available that compared the state policies on a common measurable scale before this study. This is the research gap the study addresses. The study builds a common comparative framework that analyses the competitive nature of these policies using every incentive identified across the uploaded policy documents, rather than a narrower, headline set of indicators.

2.2 Project Objectives

- To assess how well the Data Centre policies of major Indian states support investment, by examining the fiscal, regulatory, infrastructure, sustainability, and emerging technology commitments each state has made, and to allow these states to be compared on a common basis.

- To distinguish between incentives that are genuinely available to Data Centre investors and incentives that only appear to be available because they are drawn from a broader IT or industrial policy. This is intended to help state governments understand where their own policy language may be creating uncertainty for investors rather than clarity.
- To identify the specific weaknesses holding states back, so that state governments have a concrete basis for reform rather than a general sense that their policy needs improvement.
- To discourage a pattern of states competing mainly by offering larger subsidies, by showing that a well rounded policy, one that also addresses infrastructure reliability, sustainability, and ease of doing business, is a stronger and more durable basis for attracting investment than fiscal generosity alone.
- To contribute a shared reference point for what a credible state Data Centre policy should contain, at a time when India is actively considering a national policy framework for the sector.
-

2.3 Scope and Limitations

Scope

- The study is limited to officially notified state Data Centre policies relating to data centre investment. Memoranda of Understanding is not considered.
- The analysis includes states where a dedicated Data Centre policy exists, or where explicit Data Centre provisions are included within a broader IT or industrial policy.
- The focus is strictly on policy design, evaluating the commitments made on paper rather than implementation outcomes or project execution.

Limitations

- Since the study relies on publicly available documents, any policy provisions that are not disclosed publicly are outside the scope of this analysis.
- The study does not assess implementation performance, such as the actual disbursement of incentives, the pace of project execution, or the on-ground experience of investors.
- Because state policies are periodically revised or updated, the analysis reflects the policy landscape as it exists at the time of the study, and subsequent amendments may not be captured.

- Differences in drafting style, terminology and structure across states limit the comparability of the policies.
- A score of 0 is assigned to states that do not offer a given incentive, or where the incentive offered does not mention a quantified data.

3. Literature Review

3.1 Review of Relevant Studies

Industry capacity and market-sizing reports

The first set of literature comes from real estate and market-research firms that track the size and location of India's data centre capacity. JLL's India Data Center Market Dynamics report placed India's live data centre capacity at approximately 1,030 MW at the end of 2024, with a projected rise to about 1.8 GW by 2027 (JLL, 2025). CBRE's 2024 India Data Centre Market Update reported a similar figure of around 1,255 MW live by September 2024, rising to roughly 1.7 GW by 2025 (CBRE, 2024, 2025). Anarock Capital's India Data Centre Market: Growth, Trends and Outlook estimated cumulative investment of about USD 6.5 billion and capacity nearing 1.4 GW by 2024 (Anarock Capital, 2025). These reports are valuable for tracking realised investment and installed capacity by state and city. However, they generally rank locations by outcome, that is, megawatts commissioned or capital deployed, rather than by the content of the state policy that helped produce that outcome. A state can rank highly in these reports for reasons unrelated to its incentive policy, such as an existing submarine cable landing point, a power surplus, or proximity to a financial hub. This report addresses that gap by scoring policy text directly, independent of realised investment.

State-level policy comparison and legal summaries

The second strand compares specific state incentive provisions directly. The earliest and most directly comparable precedent is Cushman & Wakefield's Data Center Policies in India: A State-wise Comparison, which compared the Data Centre policies of Tamil Nadu, Uttar Pradesh, Telangana, West Bengal and Odisha, the only five states to have notified standalone Data Centre policies at that time, across power and energy infrastructure, fiscal incentives and non-fiscal incentives, and set these against the Ministry of Electronics and Information Technology's Draft Data Centre Policy of 2020 (Cushman & Wakefield, 2022). A briefing note from S&R Law, Data Centres in India: Opportunity and Incentives, later summarised stamp duty waivers, power tariff concessions and land conversion support offered by a handful of leading states (S&R Law, 2023). The Council on Energy, Environment and Water's 2025 report, Scaling India's Data Centre Ecosystem, goes further still. It compiles state-wise incentive design alongside committed investment data, and notes that only about half of the roughly USD 95 billion committed to India's data centre ecosystem between 2019 and 2025 can be traced to a specific recipient state (CEEW, 2025). These three studies are useful in flagging which incentive types exist across states, and the Cushman & Wakefield report in particular anticipated the value of a side-by-side, category-wise comparison. However, none of the three builds a common, points-based scoring instrument applied uniformly and evidenced clause-by-clause across a full national panel of states, and none extends the comparison to the 17-state panel, or the 33-incentive level of detail, used in this report. That is the specific gap this report tries to fill.

The national policy context

A third strand is the evolving national policy context against which state policies operate. India's national ambitions in this sector were first set out formally at MeitY's post-budget webinar on technology-enabled development in March 2022, at which the Ministry outlined a National Data Centre and Cloud Policy aiming to raise the country's capacity from around 500 MW to 2,500 MW within four to five years (IndiaAI, 2022). Government reporting to Parliament shows this ambition has been substantially realised and exceeded. Data centre capacity rose from about 375 MW in 2020 to approximately 1,500 to 1,575 MW by 2025, and the Ministry of Power now estimates that electricity demand from data centres could reach 13.56 GW by 2031-32, a projection that has informed measures such as the Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India Act to support future power supply for the sector (PIB, 2026; Down To Earth, 2026, reporting the same parliamentary reply). This scale of projected demand growth is directly relevant to the Infrastructure Assurance dimension used in this report, which scores whether state policies commit to assured or dual-grid power and water supply rather than leaving these commitments open-ended. Building on this trajectory, the Ministry's more recent Draft National Data Centre Policy, released for public consultation and still awaiting Cabinet and Ministry of Finance approval at the time of writing, proposes measures including conditional tax exemptions of up to 20 years, a 100 percent electricity duty exemption, input tax credit on HVAC and electrical equipment, single-window clearance, and dedicated Data Centre Economic Zones (MeitY, Draft National Data Centre Policy, 2025-26). Industry commentary on the draft has generally welcomed its intent while flagging execution risk, particularly around inter-ministerial approval and coordination with the state-level incentive frameworks scored in this report (w.media, 2026). This report treats the national draft policy, and the underlying capacity and power-demand data reported to Parliament, as context rather than as scored documents, since neither is a state-level policy text of the kind defined in Section 4.2.

Fiscal competition and investment-incentive theory

A fourth strand is the wider academic and policy literature on inter-jurisdictional competition for investment through fiscal incentives. Foundational fiscal federalism theory frames tax and public-service competition between jurisdictions as potentially efficient when residents or capital are mobile (Tiebout, 1956). A large body of subsequent tax-competition literature is more cautious. It warns of a race to the bottom, in which competing jurisdictions offer progressively larger incentives that erode the tax base without necessarily increasing net investment (UNESCAP, undated survey; Columbia Center on Sustainable Investment, 2022). Reviewing global evidence for the World Bank, James (2013) finds that fiscal incentives generally have only a modest independent effect on investment-location decisions once a jurisdiction's underlying investment climate is accounted for, an argument the OECD had made earlier in its own review of investment tax incentives (OECD, 2008, as summarised in James, 2013). In the Indian context, this literature is echoed in Invest India's own account of the shift toward competitive federalism among states since 2015, and in the Department for Promotion of Industry and Internal Trade's state-level

Business Reform Action Plan, which itself scores states on regulatory and procedural reform (Invest India, 2026; DPIIT, Business Reform Action Plan). This report's Regulatory / Procedural Incentives dimension draws on the same underlying idea: that procedural ease is a distinct and measurable form of state competitiveness, separate from headline fiscal generosity.

Composite-index construction methodology

The fifth strand concerns how to build and stress-test a composite score once individual indicators have been collected. The OECD and Joint Research Centre of the European Commission's Handbook on Constructing Composite Indicators sets out established practice for combining multiple sub-indicators into a single index. It places particular emphasis on testing a composite ranking's sensitivity to the weights assigned to its components, since a ranking built on a single, undisclosed weighting scheme can be misleading (OECD/JRC, 2008). The Monte Carlo weighting-sensitivity simulation described in Section 4.3 of this report applies that principle directly. It treats the relative importance of the five policy dimensions as uncertain rather than fixed, and reports each state's rank as a distribution across 20,000 trials rather than as a single number.

Taken together, this report differs from the existing literature in three respects. First, it scores every state against the same 33-incentive rubric, applied uniformly and drawn only from notified policy text, so that scores are directly comparable rather than resting on a narrative comparison of a handful of headline provisions, as in the market and legal-summary reports reviewed above. Second, it explicitly separates Data-Centre-specific eligibility from generic IT/ITeS eligibility, addressing a common source of over-statement in industry comparisons that read a state's general IT incentives as automatically available to Data Centres. Third, it applies the composite-index literature's emphasis on weighting-sensitivity testing, which is well established in general index-construction methodology but is not yet common practice in industry Data-Centre-policy comparisons.

3.2 Key Concepts and Definitions

- Scorecard: 33 individual incentives, grouped into five dimensions, Fiscal / Financial Incentives, Regulatory / Procedural Incentives, Infrastructure Assurance, Sustainability and Green Incentives, and Emerging / Other Incentives. Each incentive is scored 0 to 4, for a maximum raw score of 132.
- Strict Data-Centre Applicability Rule: an incentive is scored only where the published policy document explicitly makes Data Centres eligible. A generic IT/ITeS/BPO/ESDM incentive that does not name Data Centres is scored 0 for that incentive, even where a related general policy is generous.
- 0-4 Scoring Band: 0 means not provided or mentioned but not in a quantified manner. A score of 1 to 4 means the incentive is provided, with the band, 1 for limited, 2 for

moderate, 3 for strong, and 4 for very strong or comprehensive, set by a reference rubric specific to each incentive.

- Composite Policy Index: a 0 to 100 index built by averaging each state's raw score within each of the five dimensions, weighting the five dimension averages equally at 20 percent, and scaling the result by a factor of 5. States are then classified as Leading (80-100), Established (60-79), Emerging (40-59), Nascent (20-39), or Not Yet Developed (0-19).
- Equal-Weight Rank versus Mean Rank (Monte Carlo): the Equal-Weight Rank orders states when every dimension is weighted equally, that is, the rank implied by the Composite Policy Index. The Mean Rank is the average rank of a state across 20,000 trials in which the five dimension weights are drawn randomly.
- Swing Factor: for a given state, the dimension whose weight is most positively correlated with that state's rank improving, or with that state's rank worsening, across the simulated trials.

4. Methodology

4.1 Research Design

The SDPCI follows a two-stage research design. The first stage is a structured, rubric-based scoring exercise. Each state is scored against 33 incentives, grouped into 5 dimensions, on a common 0 to 4 scale, and these scores are combined, under an equal-weighting default, into a single Composite Policy Index used to rank and classify all 17 states. The second stage is a Monte Carlo robustness check. Rather than treat the equal-weighting scheme as the only valid basis for ranking states, 20,000 alternative weighting schemes are simulated, and the resulting distribution of ranks is used to separate states whose position is structurally solid from states whose position depends heavily on the specific weights chosen.

This two-stage design was chosen deliberately. A single equal-weighted ranking is easy to communicate, but it hides an important assumption, that fiscal incentives, regulatory ease, infrastructure assurance, sustainability, and emerging support all matter equally to an investor deciding where to locate a data centre. In practice, different stakeholders are likely to weigh these dimensions differently. A finance-focused reader may care more about fiscal incentives, while an operations-focused reader may care more about power and water assurance. Rather than defend one fixed weighting scheme as correct, the research design treats the weighting choice itself as an open question and tests it directly, so that the report can tell readers which parts of the ranking hold up regardless of that choice, and which parts are sensitive to it.

The unit of analysis throughout is the state. Scores are not built up from individual projects or individual companies. Each state's policy text, or the relevant policy chapter where the state does not have a standalone Data Centre policy, is read as a single document and scored incentive by incentive. This keeps the exercise comparable across very different policy instruments, from a dedicated, multi-year data centre policy with numbered clauses, to a short chapter inside a general industrial policy.

4.2 Data Collection Methods

Indicator scores were built up from official government orders and gazette notifications wherever these could be located. Examples include Andhra Pradesh's G.O.Ms.No.9 dated 12 December 2024, Haryana's New Data Centre Policy 2026 gazette notification, and the respective notified policy texts of Odisha, Uttar Pradesh, West Bengal and Rajasthan.

Data collection followed a consistent sequence for every state. First, a search was carried out for the state's official gazette notification or government order establishing a dedicated Data Centre policy. Second, where no dedicated instrument existed, the state's broader industrial, IT or ITeS policy was checked for data-centre-specific clauses, such as named incentive rates for data centre projects or a dedicated chapter. Third, each incentive was scored and cross-checked.

Each of the 33 incentives was scored from 0 to 4 and grouped into the five dimensions summarised in Table 4.1. Fiscal / Financial Incentives, the largest dimension, has 15 incentives; Regulatory / Procedural Incentives has 7; Infrastructure Assurance has 3; Sustainability and Green Incentives has 6; and Emerging / Other Incentives has 2, for a combined maximum of 132 points.

Dimension	Number of incentives	Maximum raw points
Fiscal / Financial Incentives	15	60
Regulatory / Procedural Incentives	7	28
Infrastructure Assurance	3	12
Sustainability and Green Incentives	6	24
Emerging / Other Incentives	2	8
Total	33	132

Table 4.1: The 33-incentive Scorecard, by dimension

4.3 Data Analysis Techniques

Composite scoring. For each state, the five dimension-level averages, each on a 0 to 4 scale, are combined using the default equal weight of 20 percent per dimension, and the result is scaled by a factor of 5 to produce a Composite Policy Index on a 0 to 100 scale. States are then placed into four bands based on this score, Established (60 to 79), Emerging (40 to 59), Nascent (20 to 39), and Not Yet Developed (0 to 19). No state scored in the top band, Leading (80 to 100), or in the Established band, in this round. The scaling factor of 5 follows directly from the arithmetic, five dimensions, each averaging up to 4 points, sum to a maximum of 20 points, and multiplying by 5 rescales that maximum to 100.

To test how much these rankings depend on the equal-weighting assumption, 20,000 alternative weight vectors, each with five non-negative components summing to 1, were drawn from a Dirichlet(1,1,1,1,1) distribution. This distribution is uniform over every possible weighting scheme and does not favour any one dimension over another, which makes it a neutral way to explore the full space of reasonable weighting choices rather than a small, hand-picked set of alternatives. For each of the 20,000 draws, a full composite score and rank was recalculated for all 17 states, producing a distribution of 20,000 possible ranks for every state. The spread of this distribution, shown as a boxplot in Figure 5.1 in Section 5.2, shows how rank-stable each state is. A narrow box means the state holds roughly the same rank no matter which weighting scheme is used. A wide box means its rank depends heavily on that choice.

A Dirichlet distribution was chosen over simpler alternatives, such as varying one dimension at a time while holding the others fixed, because it captures the full range of ways five weights can trade off against each other simultaneously, while still guaranteeing that every draw sums to 100

percent. This is closer to how a real disagreement about weighting would look in practice, raising the importance of one dimension necessarily lowers the combined importance of the rest, and the Dirichlet draws respects that constraint on every trial.

Driver, or swing-factor, analysis. For each state, the Pearson correlation was calculated between that state's simulated rank across the 20,000 trials and each dimension's weight in the corresponding trial. Because a lower rank number means a better position, a negative correlation means that increasing a dimension's weight improves that state's rank, and a positive correlation means that increasing the weight makes the rank worse. The dimension with the most negative correlation for a given state is reported as that state's strongest upward swing factor, and the dimension with the most positive correlation is reported as its strongest downward swing factor (Section 5.3, Table 5.4). The same correlation matrix was also read dimension by dimension, to identify, for each of the five dimensions, which state stands to gain the most and which stands to lose the most if that dimension were given more weight.

5. Findings and Analysis

5.1 Data Presentation

Table 5.1 presents the full 17-state scorecard, ranked by the Composite Policy Index, alongside the score, out of 132, and the resulting policy-maturity category. Four states reach the Emerging band, Rajasthan, Uttar Pradesh, Odisha and Gujarat. Ten states fall in the Nascent band. Three states, Andhra Pradesh, Bihar and Kerala, fall in the Not Yet Developed band. No state reaches the Established or Leading bands.

Rank	State	Composite Policy Index (0–100)	Score (out of 132)	Category
1	Rajasthan	56.1	66 / 132	Emerging
2	Uttar Pradesh	43.3	52 / 132	Emerging
3	Odisha	41.8	58 / 132	Emerging
4	Gujarat	41.1	52 / 132	Emerging
5	Karnataka	38.7	43 / 132	Nascent
6	Madhya Pradesh	37.3	51 / 132	Nascent
7	Punjab	36.2	47 / 132	Nascent
8	Maharashtra	34.3	46 / 132	Nascent
9	Telangana	27.9	31 / 132	Nascent
10	Haryana	27.6	35 / 132	Nascent
11	West Bengal	26.9	31 / 132	Nascent
12	Chhattisgarh	26.5	48 / 132	Nascent
13	Tripura	23.9	31 / 132	Nascent
14	Jharkhand	20.6	34 / 132	Nascent
15	Andhra Pradesh	18.4	23 / 132	Not Yet Developed
16	Bihar	10.0	20 / 132	Not Yet Developed
17	Kerala	4.7	5 / 132	Not Yet Developed

Table 5.1: State Data Centre Policy Index, ranked by Composite Policy Index (equal-weight scheme)

Table 5.2 breaks each state's score into its five constituent dimensions, revealing that raw totals can mask very different policy profiles. Chhattisgarh's score of 48 out of 132, for instance, is driven almost entirely by a Fiscal / Financial score of 35 out of 60, the highest Fiscal score of any state, while its Infrastructure Assurance score is 0 out of 12. Rajasthan, by contrast, scores well across

all five dimensions (24 Fiscal, 16 Regulatory, 10 Infrastructure Assurance, 12 Sustainability, 4 Emerging), which is the underlying reason it proves to be the most rank-stable state in the Monte Carlo analysis in Section 5.2.

State	Fiscal /60	Regulatory /28	Infra. Assur. /12	Sustainability /24	Emerging /8
Rajasthan	24	16	10	12	4
Uttar Pradesh	20	14	8	7	3
Odisha	30	13	6	6	3
Gujarat	24	9	9	8	2
Karnataka	14	15	8	3	3
Madhya Pradesh	25	16	5	2	3
Punjab	21	14	7	3	2
Maharashtra	19	17	5	3	2
Telangana	8	12	9	2	0
Haryana	16	9	9	1	0
West Bengal	10	12	9	0	0
Chhattisgarh	35	8	0	2	3
Tripura	12	8	6	5	0
Jharkhand	20	9	4	1	0
Andhra Pradesh	12	5	5	0	1
Bihar	16	3	0	0	1
Kerala	2	1	2	0	0

Table 5.2: Dimension-wise score break-up, out of the respective dimension maxima, ranked by Composite Policy Index

5.2 Analysis of Results: Monte Carlo Rank-Robustness

To test whether the ranking in Table 5.1 is robust to the analyst's choice of dimension weights, the SDPCI Monte Carlo Driver Analysis re-ran the ranking 20,000 times, each time drawing a different random weighting across the five dimensions. Table 5.3 reports, for each state, the Equal-Weight Rank, the Mean and Median Rank across the 20,000 trials, the Best and Worst rank the state achieved in any trial, and the percentage of trials in which the state finished in the Top-3 and Top-5.

State	Equal-Wt Rank	Mean Rank	Median Rank	Best	Worst	% Top-3	% Top-5
Rajasthan	1	1	1	1	4	100.0%	100.0%
Uttar Pradesh	2	3	3	2	7	73.0%	96.3%
Odisha	3	3.9	4	1	11	46.4%	85.3%
Gujarat	4	4.5	4	2	11	42.9%	68.3%
Karnataka	5	5.4	5	2	11	14.9%	61.8%
Madhya Pradesh	6	6.2	6	2	12	11.5%	44.2%
Punjab	7	6.9	7	4	9	0.0%	5.2%
Maharashtra	8	7.9	8	1	12	4.3%	13.1%
Telangana	9	10.2	10	2	16	0.4%	6.3%
Haryana	10	10.4	11	3	16	0.5%	3.7%
West Bengal	11	11.1	11	2	16	0.2%	2.0%
Chhattisgarh	12	10.3	10	1	16	5.9%	13.1%
Tripura	13	12.1	12	5	16	0.0%	0.7%
Jharkhand	14	13.3	14	8	16	0.0%	0.0%
Andhra Pradesh	15	14.1	15	10	16	0.0%	0.0%
Bihar	16	15.8	16	10	17	0.0%	0.0%
Kerala	17	16.8	17	15	17	0.0%	0.0%

Table 5.3: Rank stability across 20,000 randomly weighted dimension schemes

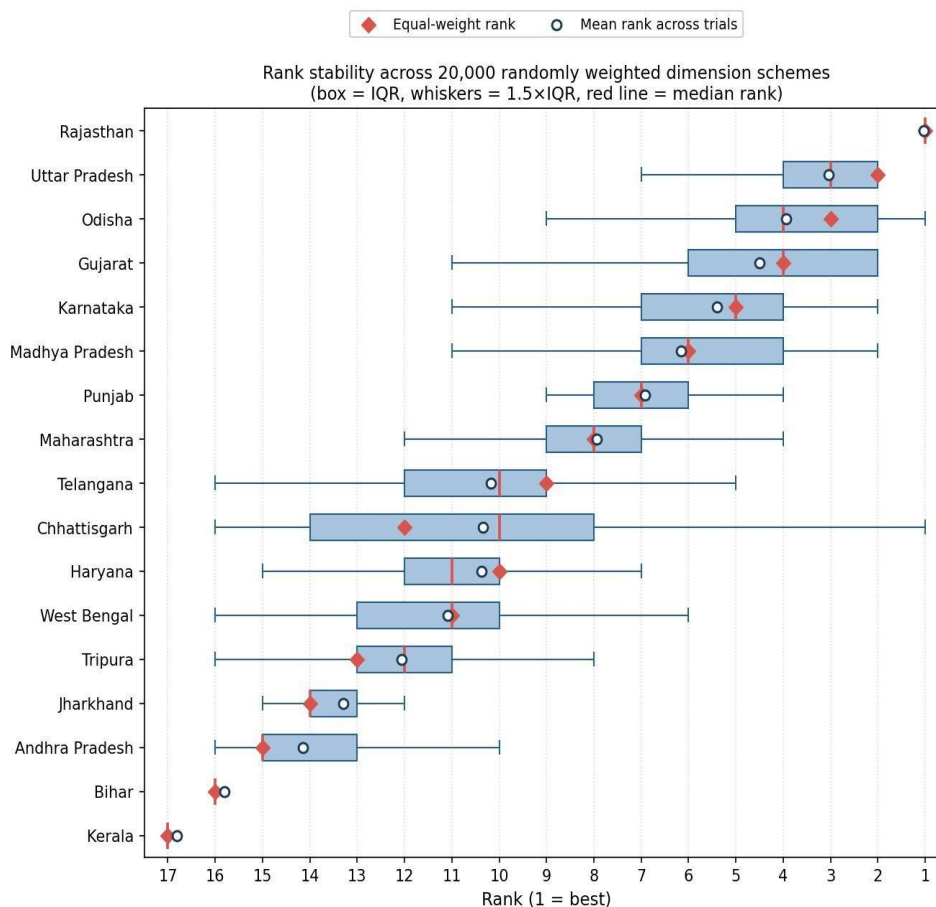


Figure 5.1: Rank stability across 20,000 randomly weighted dimension schemes. Box = interquartile range, whiskers = 1.5×IQR, red line = median rank, diamond = equal-weight rank, circle = mean rank across trials.

Table 5.4 lists, for each state, the dimension most positively associated with its rank improving (swing-up) and the dimension most positively associated with its rank worsening (swing-down).

State	Dimension pulling rank up	Corr.	Dimension pulling rank down	Corr.
Rajasthan	Infrastructure Assurance	-0.12	Fiscal / Financial Incentives	0.37
Uttar Pradesh	Emerging / Other Incentives	-0.25	Fiscal / Financial Incentives	0.69
Odisha	Fiscal / Financial Incentives	-0.57	Infrastructure Assurance	0.71
Gujarat	Infrastructure Assurance	-0.53	Regulatory / Procedural Incentives	0.71

State	Dimension pulling rank up	Corr.	Dimension pulling rank down	Corr.
Karnataka	Infrastructure Assurance	-0.37	Fiscal / Financial Incentives	0.80
Madhya Pradesh	Regulatory / Procedural Incentives	-0.42	Infrastructure Assurance	0.76
Punjab	Infrastructure Assurance	-0.13	Emerging / Other Incentives	0.33
Maharashtra	Regulatory / Procedural Incentives	-0.77	Infrastructure Assurance	0.68
Telangana	Infrastructure Assurance	-0.74	Fiscal / Financial Incentives	0.67
Haryana	Infrastructure Assurance	-0.85	Emerging / Other Incentives	0.37
West Bengal	Infrastructure Assurance	-0.79	Sustainability & Green Incentives	0.43
Chhattisgarh	Fiscal / Financial Incentives	-0.62	Infrastructure Assurance	0.75
Tripura	Sustainability & Green Incentives	-0.83	Emerging / Other Incentives	0.36
Jharkhand	Fiscal / Financial Incentives	-0.68	Emerging / Other Incentives	0.38
Andhra Pradesh	Emerging / Other Incentives	-0.64	Regulatory / Procedural Incentives	0.39
Bihar	Emerging / Other Incentives	-0.43	Infrastructure Assurance	0.51
Kerala	Infrastructure Assurance	-0.74	Fiscal / Financial Incentives	0.32

Table 5.4: Swing-factor dimensions by state

5.3 Discussion of Key Findings

Policy balance is a stronger predictor of a stable rank than incentive size. The Monte Carlo simulation shows that states with a broad, even spread of scores across the five dimensions hold their position far more reliably than states with a large score concentrated in one dimension. Rajasthan is the clearest example. It is the only state that lands in the Top 4 in every one of the 20,000 trials, with a Mean Rank of 1.0 that is identical to its Equal Weight Rank of 1.

Its swing down dimension, Infrastructure Assurance, has a correlation of only negative 0.12, so even trials that reduce the weight of its weaker dimension rarely move it out of contention. This stability comes from its balanced profile across Fiscal, Regulatory, Infrastructure Assurance, Sustainability and Emerging incentives, combined with the highest score in the panel at 66 out of 132.

Infrastructure Assurance is the most common and most damaging weak point. Infrastructure Assurance, covering assured power, dual grid connectivity and assured water supply, is named as the swing down, rank worsening dimension for nine of the seventeen states, including Gujarat, Haryana, Karnataka, Maharashtra, Madhya Pradesh, Odisha, Punjab, Telangana and West Bengal. It also has the widest correlation range of any dimension, at 1.62, meaning it produces the largest swings in state fortunes of any of the five dimensions when its weight changes. Chhattisgarh illustrates the effect most sharply. It ranks sixth on raw score but falls to twelfth on the Composite Policy Index, a swing of six ranks, because its Infrastructure Assurance score is 0 out of 12 despite a strong Fiscal score of 35 out of 60.

Wide variation between states points to the absence of a shared policy baseline. The scorecard results show large differences not only in total score but in which dimensions each state chooses to address. Odisha concentrates its strength in Fiscal incentives, at 30 out of 60, while leaving Regulatory and Infrastructure Assurance comparatively thin. Uttar Pradesh spreads its score more evenly across all five dimensions. This unevenness across the panel, rather than a consistent structure that all states follow, is what allows dimension weighting to change state rankings as much as it does in the simulation.

The strict applicability rule shows that general IT incentives cannot be assumed to cover Data Centres.

Because the scoring credits an incentive only where the policy text explicitly names Data Centres, states with generous general IT or ITeS incentives can still score low on individual items. Punjab's general IT Policy chapter offers a 100 percent Net SGST reimbursement and an EPF and ESI reimbursement to IT, ITeS, ESDM and BPO units, but because the Data Centre chapter of its 2026 policy does not extend these to Data Centres, both are scored 0. Jharkhand's general recruitment assistance and land subsidy provisions are Data Centre eligible in places, but its EPF, R&D and Quality Certification incentives are scored 0 for the same reason. This confirms that a headline reading of a state's general policy can overstate what a Data Centre investor can actually claim.

States with unquantified incentives score poorly even where a policy exists. Kerala's IT Policy 2026 references a proposed Data Centre park regime around the Vizhinjam and DP World corridor, but leaves most Data Centre specific incentive amounts, including capital subsidy, power tariff and land subsidy percentage, unstated. This is reflected in scores of 0 across

most of the 33 incentives measured, and contributes to Kerala's position at the bottom of the panel, with a Composite Policy Index of 4.7 and a 0 percent probability of a Top 5 finish across all 20,000 trials. This shows that a stated intention to support Data Centres is not treated as equivalent to a quantified commitment.

Sustainability is comparatively underdeveloped across the panel. Sustainability scores are low relative to Fiscal and Regulatory scores for most states, including several in the Emerging band. Chhattisgarh's Sustainability score is 2 out of 24 despite its strong Fiscal position, and Andhra Pradesh has no identified Sustainability incentive for Data Centres at all. Given that Fiscal incentives dominate most states' scores while Sustainability lags behind, the panel as a whole shows an imbalance that mirrors the Infrastructure Assurance gap, though it is less severe.

The study measures policy design, not delivery. Every finding in this study is based on the text of published policy documents, not on disbursement records, approval timelines or investor experience. This is a defined limitation of the research design rather than a finding in itself, but it directly explains why no claim in this section is made about how well any state actually implements its policy commitments, only about how those commitments are written.

The Composite Policy Index reflects promised policy, not achieved outcomes. Because the index is built entirely from published incentive text, a state's score reflects what has been committed on paper. Rajasthan's leading position and the Not Yet Developed status of Andhra Pradesh, Bihar and Kerala are both statements about policy design quality, not about which state has attracted the most investment or commissioned the most capacity. This distinction is central to interpreting every result in this study correctly.

Fiscal strength alone does not produce a strong or stable overall position. Chhattisgarh has the highest Fiscal and Financial score in the panel, at 35 out of 60, yet falls six ranks once dimension weights are varied, and Karnataka reaches a higher Composite Policy Index rank than its raw score would suggest despite a comparatively modest Fiscal score of 14 out of 60. Karnataka's most exposed dimension is Fiscal and Financial Incentives, with a correlation of 0.80, the strongest positive fiscal correlation in the panel, meaning its position would improve most if fiscal weight were reduced further. Together these results show that fiscal generosity, on its own, is an unreliable basis for a strong overall ranking..

6.

Suggestions

Design principles for state policy

States should treat policy balance as more important than the size of any single incentive. The findings show that states with an even spread of commitments across fiscal, regulatory, infrastructure, sustainability, and emerging technology areas hold a stronger and more stable position than states that rely heavily on one dimension alone.

Sustainability should be treated as a core part of Data Centre policy rather than an optional addition. Since Sustainability is comparatively underdeveloped across most states studied, and since Data Centres place a significant demand on power and water resources, this dimension deserves the same level of attention as fiscal incentives.

Specific gaps that can be addressed directly

Infrastructure Assurance should be prioritised ahead of other reforms. This is the most common weakness identified across the states studied, and addressing it, through clear commitments on power supply, dual grid connectivity, and water availability, would improve the competitive position of more states than any other single change.

States should state clearly, within their Data Centre policy documents, which incentives apply to Data Centres. Several states offer strong general IT or industrial incentives that do not extend to Data Centres in practice, simply because the policy text does not say so. Making this explicit would remove unnecessary uncertainty for investors.

States should replace general statements of intent with specific, quantified commitments. A policy that states an exact subsidy amount, tariff rate, or approval timeline is more useful to an investor, and easier to compare with other states, than a policy that only expresses support in general terms.

Structural proposals

A shared minimum framework for state Data Centre policy would help reduce the wide variation currently seen across states. Such a framework could require every state to address each of the five policy dimensions, while still allowing individual states to decide the scale of their own incentives.

States could consider linking a portion of their incentives to measurable outcomes, such as actual investment made, capacity commissioned, or renewable energy adopted. This would reduce reliance on unconditional subsidy competition and encourage incentives that reward genuine performance rather than policy design alone.

Areas for further work

This study assesses what state policies commit to, not how well those commitments are delivered. A useful next step would be to examine implementation directly, including approval timelines, disbursement of incentives, and the experience of investors who have already set up operations under these policies.

States may also wish to consider whether they can differentiate themselves by the type of Data Centre activity they are best suited to support, such as hyperscale facilities, edge infrastructure, or renewable powered facilities. This was not something the current study was designed to measure, but it may be a useful direction for individual states to explore further.

Finally, a future assessment could build on this study by measuring policy effectiveness rather than policy design alone, looking at outcomes such as realised investment and commissioned capacity rather than the content of published policy documents.

7. Learnings from Internship

Working on the State Data Centre Policy Index gave practical exposure to building a policy-comparison instrument from primary government documents rather than secondary commentary, and to the discipline required to score consistently across states with very different drafting styles and levels of detail. Reading gazette notifications, government orders and policy chapters side by side, and applying the same strict Data-Centre-applicability rule to each of them, was a useful lesson in how much a headline reading of a policy can overstate its actual, claimable benefit.

The assignment also provided direct experience with sensitivity and robustness testing of a composite index, using a Monte Carlo simulation over dimension weights, and with translating a large body of numerical results into swing-factor findings that a non-technical policy audience can act on. This reinforced the value of testing an assumption, in this case equal weighting across dimensions, rather than presenting it as the only possible basis for ranking states, and of separating a state's raw score from the robustness of its position under alternative weighting schemes.

More generally, the assignment deepened an understanding of competitive federalism in the Indian context, of how state governments design and revise investment-incentive policy in response to one another, and of the practical constraints, incomplete public disclosure, inconsistent terminology, and periodic revision, that make any state-level policy comparison an ongoing rather than a one-time exercise.

8. References

- Anarock Capital. (2025). *India data centre market: Growth, trends and outlook*.
- CBRE. (2024). *India data centre market update*.
- CBRE. (2025). *India data centre market update*.
- Columbia Center on Sustainable Investment. (2022). *Tax incentives and the race to the bottom in investment competition*.
- Council on Energy, Environment and Water. (2025). *Scaling India's data centre ecosystem*.
- Cushman & Wakefield. (2022). *Data centre policies in India: A state-wise comparison*.
- Department for Promotion of Industry and Internal Trade. (n.d.). *Business reform action plan*.
- Down To Earth. (2026). *Reporting on parliamentary reply on data centre power demand*.
- IndiaAI. (2022). *MeitY post-budget webinar on technology-enabled development, national data centre and cloud policy*.
- Invest India. (2026). *Competitive federalism among Indian states since 2015*.
- James, S. (2013). *Tax and non-tax incentives and investments: Evidence and policy implications*. World Bank.
- JLL. (2025). *India data centre market dynamics*.
- Ministry of Electronics and Information Technology. (2025-2026). *Draft national data centre policy*.
- OECD. (2008a). *Tax effects on foreign direct investment*.
- OECD, & Joint Research Centre, European Commission. (2008b). *Handbook on constructing composite indicators: Methodology and user guide*.
- Press Information Bureau, Government of India. (2026). *Growth of data centre capacity in India*. <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2239616®=3&lang=1>
- S&R Law. (2023). *Data centres in India: Opportunity and incentives*.
- Tiebout, C. M. (1956). A pure theory of local expenditures. *Journal of Political Economy*, 64(5), 416-424.

UNESCAP. (n.d.). *Survey on tax incentives and investment competition*.

w.media. (2026). *Industry commentary on the draft national data centre policy*.

State government sources. Official Data Centre, IT, or Industrial policy documents of Andhra Pradesh, Bihar, Chhattisgarh, Gujarat, Haryana, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Punjab, Rajasthan, Telangana, Tripura, Uttar Pradesh, and West Bengal, listed in full in Appendix 8.3.

9. Appendices

9.1 Survey Instruments

This study is a desk-based document-scoring exercise rather than a survey. No primary survey instrument, questionnaire or interview schedule was used. All scores in Section 5 were derived directly from the official policy documents listed in Appendix 8.3, supplemented where noted in Section 4.2.

9.2 Raw Data

The table below reproduces the complete 33-incentive raw score matrix for all 17 states, grouped by dimension, on the 0 to 4 scoring band described in Section 3.2. State names are abbreviated for space: AP (Andhra Pradesh), GJ (Gujarat), HR (Haryana), KA (Karnataka), CG (Chhattisgarh), MH (Maharashtra), MP (Madhya Pradesh), OD (Odisha), RJ (Rajasthan), PB (Punjab), JH (Jharkhand), TG (Telangana), WB (West Bengal), BR (Bihar), KL (Kerala), TR (Tripura), UP (Uttar Pradesh). Columns are ordered by Composite Policy Index rank, as in Table 5.1.

Incentive	R J	U P	O D	G J	K A	M P	P B	M H	T G	H R	W B	C G	T R	J H	A P	B R	K L
FISCAL / FINANCIAL INCENTIVES																	
Capital (CAPEX) Subsidy on Fixed Investment	3	1	3	1	1	4	3	0	0	0	0	4	4	3	2	4	0
Interest Subsidy on Term Loan	3	3	3	3	0	0	0	0	0	0	0	2	0	2	0	3	0
Power Tariff Subsidy / Concessional Tariff	1	4	3	3	2	3	1	2	1	2	1	0	1	2	1	2	0
Electricity Duty Exemption	2	3	3	4	2	3	3	4	0	4	0	3	0	2	0	1	1
Stamp Duty Exemption / Concession	3	3	3	4	4	3	3	4	0	3	4	4	0	3	3	0	0
Registration Fee Exemption / Concession	0	0	0	4	2	3	0	0	0	0	4	2	0	3	0	0	0
Land Cost Subsidy / Rebate	3	3	2	0	2	2	1	1	1	0	0	0	1	3	1	0	0
Priority / Long-Term Subsidised Land Allotment	2	3	2	1	1	2	2	2	2	1	1	1	3	1	1	0	1
SGST / Net-SGST Reimbursement	3	0	2	4	0	0	3	0	0	3	0	4	0	0	4	0	0
Employment Generation Incentive (cash per employee)	0	0	2	0	0	0	2	1	0	3	0	3	0	1	0	0	0
EPF (Provident Fund) Reimbursement	0	0	3	0	0	0	0	0	0	0	0	3	0	0	0	3	0
Lease Rental Subsidy (for data centre units)	0	0	0	0	0	2	0	1	2	0	0	3	3	0	0	3	0
R&D Expenditure Reimbursement	2	0	1	0	0	2	1	2	2	0	0	3	0	0	0	0	0

Incentive	R J	U P	O D	G J	K A	M P	P B	M H	T G	H R	W B	C G	T R	J H	A P	B R	K L
Skill Development / Training Cost Reimbursement	2	0	3	0	0	1	2	2	0	0	0	2	0	0	0	0	0
Transportation of Plant & Machinery Reimbursement	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
REGULATORY / PROCEDURAL INCENTIVES																	
Single-Window Clearance	2	2	2	2	3	3	4	4	2	2	2	3	2	3	2	2	1
“Essential Service” Status Designation	3	2	4	2	2	3	0	4	4	2	4	0	2	3	0	0	0
Relaxation in Building Norms / FSI	4	2	0	4	4	3	3	3	2	3	4	0	2	2	2	0	0
Self-Certification for Regulatory Compliance	3	4	2	1	2	1	3	1	3	0	2	1	2	0	1	1	0
Exemption from Select Labour-Law Provisions	2	4	3	0	2	2	0	0	0	0	0	0	0	0	0	0	0
Quality Certification Reimbursement (ISO / ISMS etc.)	0	0	0	0	0	2	0	2	0	0	0	2	0	0	0	0	0
Time-Bound Approvals	2	0	2	0	2	2	4	3	1	2	0	2	0	1	0	0	0
INFRASTRUCTURE ASSURANCE																	
Assured / Uninterrupted Power Supply	4	0	3	3	2	2	3	3	4	3	4	0	0	2	2	0	2
Dual Power Grid Connectivity	3	4	0	3	3	2	3	1	2	3	2	0	4	2	2	0	0
Assured / Uninterrupted Water Supply	3	4	3	3	3	1	1	1	3	3	3	0	2	0	1	0	0
SUSTAINABILITY & GREEN INCENTIVES																	
Renewable Energy Open Access / Procurement	4	3	4	2	1	2	2	3	1	0	0	0	2	1	0	0	0
Incentive for Renewable Energy Adoption (green power tariff benefit)	3	4	0	3	2	0	1	0	1	0	0	0	1	0	0	0	0
Green Building / IGBC Certification Reimbursement	1	0	1	0	0	0	0	0	0	0	0	0	2	0	0	0	0
Effluent Treatment Equipment Reimbursement	3	0	1	2	0	0	0	0	0	0	0	1	0	0	0	0	0
Power Tariff Reimbursement for Effluent Treatment Operations	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Rainwater Harvesting System Reimbursement	1	0	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0
EMERGING / OTHER INCENTIVES																	

Incentive	R J	U P	O D	G J	K A	M P	P B	M H	T G	H R	W B	C G	T R	J H	A P	B R	K L
AI / GPU Infrastructure (“Compute Booster”) Incentive	2	0	2	1	1	1	1	1	0	0	0	0	0	0	1	0	0
Additional Incentives for Backward / Underdeveloped Regions	2	3	1	1	2	2	1	1	0	0	0	3	0	0	0	1	0

Table 8.1: Full 33-incentive raw score matrix, 0-4 scale, all 17 states

9.3 Additional Supporting Documents

Table 8.2 lists the official policy document used as the primary source for each state's score. Where the primary document could not be located, or was too recent for full operating guidelines to be published, this was supplemented as described in Section 4.2.

State	Official policy document used for scoring
Rajasthan	Rajasthan Data Centre Policy 2025
Uttar Pradesh	Uttar Pradesh Data Centre Policy 2021
Odisha	Odisha Data Centre Policy (notified policy text)
Gujarat	Gujarat Data Centre Policy 2026-29
Karnataka	Karnataka Data Centre Policy document
Madhya Pradesh	Madhya Pradesh Draft IT Policy
Punjab	Punjab IT, ITeS, Data Centre and GCC Policy 2026
Maharashtra	Maharashtra IT/ITeS Policy 2023
Telangana	Telangana Data Centres Policy
Haryana	Haryana Data Centre Policy (New Data Centre Policy 2026, gazette notification)
West Bengal	West Bengal Data Centre Policy 2021
Chhattisgarh	Chhattisgarh Industrial Policy 2024-30
Tripura	Tripura Data Centre Policy 2021
Jharkhand	Jharkhand IT, Data Centre and BPO Investment Promotion Policy 2023
Andhra Pradesh	Guidelines for Data Centre Policy (G.O.Ms.No.9, dated 12 December 2024)
Bihar	Bihar Industrial (IT) Policy 2024
Kerala	Kerala IT Policy 2026

Table 8.2: Official policy documents used for scoring, by state